

Comparing irregular sampled time series using the Jensen-distance rate

Javier E. Contreras-Reyes^{a,*}, Francisco Pérez-Galarce^b, César Ojeda^c

^a*Instituto de Matemática, Física y Estadística, Facultad de Ingeniería y Negocios,
Universidad de Las Américas, Sede Viña del Mar, 7 Norte 1348, Viña del Mar, Chile*

^b*Facultad de Ingeniería y Negocios, Universidad de Las Américas, Santiago, Chile*

^c*Escuela de Estadística, Facultad de Ingeniería, Universidad del Valle, Cali, Colombia*

Abstract

.

Keywords: Irregular time series; Jensen-distance rate; stationary processes;
Lomb–Scargle periodogram; Thomson multitaper; stellar light curves

1. Introduction

A key component in several machine learning models is the distance metric, since many algorithms rely on a measure of closeness or similarity to operate. This concept is fundamental in unsupervised learning methods such as k-means, hierarchical clustering, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN), which use distance to define clusters or neighborhood structures. Similarly, in supervised learning, algorithms such as k-Nearest Neighbors (kNN) rely directly on distance metrics to identify and compare instances in the feature space. Hence, the definition and choice of the distance metric are crucial, as they strongly influence model performance, interpretability, and computational efficiency.

In particular, defining an appropriate distance metric for light curves is challenging. These astronomical time series are often irregularly sampled, meaning that observations are not taken at uniform time intervals, and each object is observed at different epochs. Additionally, the number of observations varies across objects, and measurements may be obtained in multiple photometric bands, each with its own sampling pattern. Every observation also

*Corresponding author. Email address: jecontrr@uc.cl

carries an associated photometric uncertainty, which introduces heteroscedastic noise into the data. Furthermore, the apparent magnitude of a star depends on its distance, complicating direct comparisons between objects. These factors make it challenging to define a unique, consistent measure of similarity that accurately captures the intrinsic variability and periodicity of light curves across diverse observational conditions.

According to recent reviews, the distance metrics employed in machine learning models for both supervised and unsupervised learning are commonly derived from either hand-crafted features or deep learning embeddings. Both approaches present certain limitations. Methods based on hand-crafted predictors depend heavily on the specific set of features selected, which may not fully capture the underlying variability or temporal dynamics of the data. Conversely, approaches based on deep learning embeddings rely on the training data used to learn the representations; thus, their performance and generalization capacity are constrained by the diversity, quality, and biases of the dataset employed during model training.

We propose an ad-hoc distance metric for pulsating light curves, based on the Jensen–Distance Rate (JDR) estimator ([Contreras-Reyes 2025](#)). This proposal addresses several common challenges in the automatic analysis of photometric data, including irregular temporal sampling, varying time-series lengths, photometric measurement errors, and apparent magnitude differences due to distance or interstellar extinction. The metric is based on cross-correlation between both signals and is computed in the spectral domain, allowing the comparison of light curves independently of their apparent magnitude and with robustness to noise, making it particularly useful for catalogs with low signal-to-noise ratios. In addition, we analyze its stability under variations in the number of observations and its extension to multiple photometric bands (e.g., I and V bands from the OGLE III project), showing that it can integrate spectral information across bands and relate to physical properties such as stellar color. Finally, we discuss its computational feasibility by comparing different spectral estimators (Lomb–Scargle, Thompson multitaper with varying numbers of tapers), emphasizing the trade-off between accuracy and computational cost. This metric is presented as a novel tool that builds on recent advances in machine learning applied to astronomical time series, providing an ad hoc similarity measure suitable for both unsupervised and supervised methods such as k-means and k-NN, thereby contributing to the development of more robust and physically interpretable techniques for the analysis of variable stars.

The rest of this paper is organized as follows. Section 2 reviews related work on distance metrics applied to variable star data analysis, highlight-

ing key findings and current challenges. Section 3 introduces the theoretical background necessary to understand the proposed distance metric. Section 4 presents the new distance metric and describes the contexts in which it can be effectively incorporated. Section 5 reports the experimental results and performance evaluation. Finally, Section 6 summarizes the main conclusions and outlines directions for future research.

2. Background theory

Let $x(T_t)$ and $y(T_t)$, $t = 1, 2, \dots, n$, be two weakly stationary time series with zero mean. Process $x(T_t)$ is regularly sampled if $\Delta t = T_t - T_{t-1}$ is constant for all t . In literature, a classical estimator of spectral density for weakly stationary time series has been the Schuster periodogram; however, this lacks of important statistical properties such as bias and inconsistency (variance not decreases when n increases) (Springford et al. 2020). Moreover, Schuster periodogram has fails to detects peaks in seasonal time series if they were irregularly sampled (Δt is not constant) (Vio et al. 2013). For irregularly time series, Lomb–Scargle periodogram has been considered as follows.

2.1. Lomb–Scargle periodogram

The Lomb–Scargle periodogram (LSP) (see e.g. VanderPlas 2018; Springford et al. 2020) was derived from Least Square estimation method as a path to estimate the spectral density of a irregularly time series. It mainly requires cosine and sine components of the regression, and solve a minimization problem of the absolute error between original time series and regression. The LSP estimator is given by

$$\hat{f}_x(\omega) = \frac{1}{2} \left[\frac{\left(\sum_{t=1}^n x(T_t) \cos(\omega T_t^*) \right)^2}{\sum_{t=1}^n \cos^2(\omega T_t^*)} + \frac{\left(\sum_{t=1}^n x(T_t) \sin(\omega T_t^*) \right)^2}{\sum_{t=1}^n \sin^2(\omega T_t^*)} \right], \quad (1)$$

with frequency $\omega = 2\pi\varpi$, $\varpi > 0$, $T_t^* = T_t - \tau(\omega)$, and

$$\tau(\omega) = \frac{1}{2\omega} \arctan \left[\frac{\sum_{t=1}^n \sin(2\omega T_t)}{\sum_{t=1}^n \cos(2\omega T_t)} \right].$$

The LSP estimator has some limitations in practical astronomical applications where irregular time series were sampled, for example, Vio et al. (2013) reported that LSP provides similar results than other simplest methods such

Schuster periodogram, because these methods not allowed to identify the periodic signal frequencies given the interference between the true peaks and those due to sampling. In addition, [Springford et al. \(2020\)](#) also reported that LSP suffers with the same bias, inconsistency, and spectral leakage problems that of Schuster periodogram, thus they proposed the Thomson multitaper spectral estimator to trade off bias and variance for periodic frequencies.

2.2. Thomson multitaper spectral estimator

In this section, it is considered the tapers $w_t^{(k)}$, $t = 1, \dots, n$, $k = 1, \dots, m$, as a sequence of optimal smoothed spectral windows for Fourier analysis ([Springford et al. 2020](#); [Dodson-Robinson and Haley 2024](#)). The tapers $w_t^{(k)}$ are obtained by solving the following system

$$\sum_{r=1}^n \frac{\sin(2\pi\omega b[T_t - T_r])}{\pi[T_t - T_r]} w_r^{(k)} = \lambda^{(k)} w_t^{(k)}, \quad (2)$$

where b is the bandwidth, such that oscillations in the time series reach peaks in the spectrum estimate with full width $m = 2b$. Solution of system (2) provides m independent weighted Schuster periodograms given by

$$\tilde{f}_x^{(k)}(\omega) = \left| \sum_{t=1}^n w_t^{(k)} x(T_t) e^{-i2\pi\omega T_t} \right|, \quad (3)$$

and the multitaper spectrum estimator is the average of the m quantities $\tilde{f}_x^{(k)}(\omega)$. To obtain the Lomb–Scargle periodogram based on tapers $w_t^{(k)}$, it is defined the interpolated time series by the vector product between tapers and original irregular time series, i.e., $u_t^{(k)} = w_t^{(k)} \times x_t$, and replace each one on LSP (1). Then, the multitaper spectrum estimator based on LSP is

$$\tilde{f}_u(\omega) = \frac{1}{m} \sum_{k=1}^m \hat{f}_{u^{(k)}}(\omega). \quad (4)$$

Considering several applications in astronomy, [Springford et al. \(2020\)](#) proved that LSP estimator based on Thomson multitaper solve the statistical problems mentioned latter. In the next section, latter both methods are considered to developed the JDR between two weakly stationary signals, and show some numerical examples to compare their performances.

3. Methodology

After assessing the advantages of JDR on light curves, I propose developing a multiband kNN JDR classifier as follows: 1) Select the observations in each band using an ad hoc selection method. 2) Compute the JDR for each object in the training set (possibly adopting a hierarchical approach to reduce computational cost). 3) Automatically determine the optimal value of k . 4) Properly merge the distances computed for each band.

We proposed the JDR (Contreras-Reyes 2025) as a discrepancy measure between $x(T_t)$ and $y(T_t)$. JDR is written in terms of integrals of spectrum and cross-spectrum densities given by

$$J_\alpha(\mathcal{X}, \mathcal{Y}) = \frac{\beta}{2\pi} \int_0^{2\pi} \left\{ f_x(\omega) + f_y(\omega) - 2f_{xy}(\omega) \right\} d\omega, \quad (5)$$

with $\beta = \alpha(1 - \alpha)$, $0 < \alpha < 1$. It is possible to define the cross-spectrum between x_t and y_t (Rehfeld et al. 2011) as

$$f_{xy}(\omega) = f_x(\omega) \overline{f_y(\omega)}, \quad (6)$$

where $\overline{f_y(\omega)}$ denotes the conjugate of $f_y(\omega)$.

JDR (5) represents the average distance between both stationary processes given that is an integral with respect to frequency domain. Note that if x_t and y_t are uncorrelated, the square coherence function (see e.g. Schulz and Stattegger 1997) is zero for all ω and computation of JDR is reduced only to integrals of LSP marginals. In addition, given that JDR is based on Jensen-variance distance (Kharazmi et al. 2024), the JDR has the following properties: a) positiveness, b) symmetry, c) triangle inequality, d) invariance under rescaling and transformation.

Based on estimator (1), the estimated JDR based on LSP is straightforward:

$$\hat{J}_\alpha(\mathcal{X}, \mathcal{Y}) = \frac{\beta}{2\pi} \int_0^{2\pi} \left\{ \hat{f}_x(\omega) + \hat{f}_y(\omega) - 2\hat{f}_{xy}(\omega) \right\} d\omega. \quad (7)$$

For LSP based on Thomson multitaper (4), the estimated JDR is derived as

$$\begin{aligned} \tilde{J}_\alpha(\mathcal{U}, \mathcal{V}) &= \frac{1}{m} \sum_{k=1}^m \hat{J}_\alpha(\mathcal{U}^{(k)}, \mathcal{V}^{(k)}) \\ &= \frac{\beta}{2\pi m} \sum_{k=1}^m \int_0^{2\pi} \left\{ \hat{f}_{u^{(k)}}(\omega) + \hat{f}_{v^{(k)}}(\omega) - 2\hat{f}_{u^{(k)}v^{(k)}}(\omega) \right\} d\omega, \end{aligned} \quad (8)$$

with $v_t^{(k)} = w_t^{(k)} \times y_t$.

Estimator (7) involves the computation of only 3 integrals, while estimator (8) requires the computation of $3m = 6b$ integrals. Each integral of LSP could be quickly computed using the `integrate` function of R software (Piessens et al. 1983). This method proved to be more effective than others such as MonteCarlo one.

Considering both (7) and (8), it is possible to consider a mixed estimator that involves original x_t and tapered $u_t^{(k)}$ time series as

$$\begin{aligned}\tilde{J}_\alpha(\mathcal{X}, \mathcal{U}) &= \frac{1}{m} \sum_{k=1}^m \hat{J}_\alpha(\mathcal{X}, \mathcal{U}^{(k)}) \\ &= \frac{\beta}{2\pi} \left[\int_0^{2\pi} \hat{f}_x(\omega) d\omega + \frac{1}{m} \sum_{k=1}^m \int_0^{2\pi} \left\{ \hat{f}_{u^{(k)}}(\omega) - 2\hat{f}_{xu^{(k)}}(\omega) \right\} d\omega \right],\end{aligned}\tag{9}$$

which involves $2m + 1 = 4b + 1$ integrals. Estimator (9) not measures the distance between two irregular time series, but measures the discrepancy between an irregular time series with its tapered version, i.e., the difference between LSP and Thomson multitaper estimator, both applied to the same x_t .

4. JDR-based k -medoids clustering

Since the proposed JDR is defined as a pairwise dissimilarity between irregularly sampled light curves, a natural extension to unsupervised learning is a medoid-based clustering procedure. In contrast to k -means, which requires a centroid representation and iterative updates in a vector space—thus recomputing distances between each observation and all centroids at every iteration— k -medoids operates exclusively on pairwise dissimilarities. Consequently, it can be directly applied to the precomputed JDR matrix. In this framework, each cluster is represented by an observed light curve, selected as the medoid that minimizes the total within-cluster JDR dissimilarity.

Let K denote the number of clusters, and let $m_1, \dots, m_K \in \mathcal{N}$ be the indices of the selected medoids. Given the JDR-based dissimilarity d_{ij} between light curves i and j , the JDR k -medoids problem can be formulated as

$$\min_{m_1, \dots, m_K} \sum_{i=1}^n \min_{k=1, \dots, K} d_{im_k}.\tag{10}$$

This formulation assigns each light curve to its closest medoid and minimizes the total dissimilarity.

An equivalent clusterwise formulation is given by

$$\min_{\{C_1, \dots, C_K\}, \{m_1, \dots, m_K\}} \sum_{k=1}^K \sum_{i \in C_k} d_{im_k}, \quad (11)$$

subject to

$$m_k \in C_k, \quad C_k \cap C_\ell = \emptyset \quad (k \neq \ell), \quad \bigcup_{k=1}^K C_k = \mathcal{N}. \quad (12)$$

To solve this, we use the procedure presented in Algorithm 1.

Algorithm 1 Iterative JDR-based k -medoids

Require: Set of light curves $\mathcal{N} = \{1, \dots, n\}$, number of clusters K , JDR dissimilarity matrix $\mathbf{D} = (d_{ij})$

Ensure: Set of medoids $\{m_1, \dots, m_K\}$ and clusters $\{C_1, \dots, C_K\}$

- 1: Select an initial set of medoids $\{m_1, \dots, m_K\} \subseteq \mathcal{N}$
- 2: Set `changed` $\leftarrow$ `true`
- 3: **while** `changed` **do**
- 4: Set `changed` $\leftarrow$ `false`
- 5: **for** each $i \in \mathcal{N}$ **do** $\triangleright$ Assignment step
- 6: Assign i to its nearest medoid:

$$c(i) \leftarrow \arg \min_{k=1, \dots, K} d_{im_k}$$

- 7: **end for**
- 8: Define clusters $C_k \leftarrow \{i \in \mathcal{N} : c(i) = k\}$ for all $k = 1, \dots, K$
- 9: **for** each $k = 1, \dots, K$ **do** $\triangleright$ Medoid update step
- 10: Compute

$$m_k^* \leftarrow \arg \min_{j \in C_k} \sum_{i \in C_k} d_{ij}$$

- 11: **if** $m_k^* \neq m_k$ **then**
 - 12: $m_k \leftarrow m_k^*$
 - 13: `changed` $\leftarrow$ `true`
 - 14: **end if**
 - 15: **end for**
 - 16: **end while**
 - 17: **return** $\{m_1, \dots, m_K\}$ and $\{C_1, \dots, C_K\}$
-

5. Simulations

5.1. Irregular Autoregressive process

In this section, we consider the irregular Autoregressive (iAR) model proposed by [Eyheramendy et al. \(2018\)](#) for astronomical time series. Denote $\{y(T_t)\}$ an observation measured at an increasing sequence of observational times $\{T_t\}$, $t = 1, \dots, n$. The iAR process is defined as

$$y(T_t) = \phi^{\Delta t} y(T_{t-1}) + \varepsilon(T_t), \quad (13)$$

where $\{\varepsilon(T_t)\}$ are independent random variables such that $\mathbb{E}[\varepsilon(T_t)] = 0$ and $\mathbb{E}[\varepsilon^2(T_t)] = \sigma^2(1 - \phi^{2\Delta t})$. Authors proved that process (13) is weakly stationary and ergodic.

Based on process (13), the negative log-likelihood of iAR process can be written as

$$\ell(\theta | \{y(T_t)\}) = \frac{n}{2} \log(2\pi) + \frac{1}{2} \sum_{t=1}^n \left(\log \nu(T_t) + \frac{e^2(T_t)}{\nu(T_t)} \right), \quad (14)$$

with $\theta = (\sigma, \phi)$,

$$e(T_t) = \begin{cases} y(T_1), & t = 1, \\ y(T_t) - \phi^{\Delta t} y(T_{t-1}), & t > 1; \end{cases}$$

and

$$\nu(T_t) = \begin{cases} \sigma^2, & t = 1; \\ \sigma^2(1 - \phi^{2\Delta t}), & t > 1. \end{cases}$$

Function (14) is numerically maximized to obtain the maximum likelihood estimator (MLE) $\hat{\phi}$ based on

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^n \frac{\left(y(T_t) - \hat{y}(T_t) \right)^2}{1 - \hat{\phi}^{2\Delta t}},$$

with

$$\hat{y}(T_t) = \begin{cases} 0, & t = 1, \\ \hat{\phi}^{\Delta t} y(T_{t-1}), & t > 1. \end{cases}$$

[Eyheramendy et al. \(2018\)](#) shown that MLE $\hat{\phi}$ has an asymptotic normal distribution.

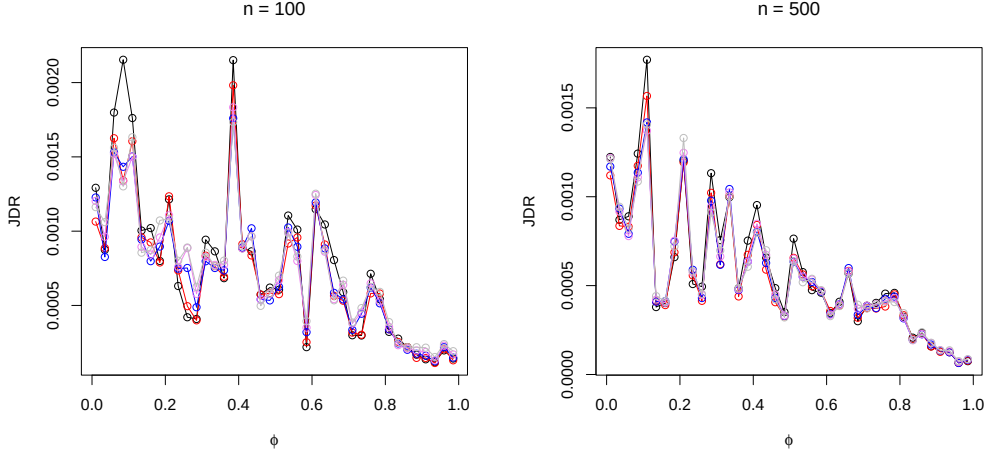


Figure 1: .

5.2. Irregular Autoregressive Moving Average process

Ojeda et al. (2021, 2023); Godoy and Ojeda (2025)

6. Light curves discoveries based on distances

Variable stars are crucial celestial objects for understanding the structure of the universe, as some of them act as standard candles for distance measurements (e.g. RR Lyrae and Cepheids). Others, such as Delta Scuti stars, play a fundamental role in asteroseismology, since their pulsations provide valuable information about the internal structure and evolutionary state of stars. The observed variability can originate from internal physical processes or external causes, such as orbiting planets or the presence of binary or multiple systems. A particular case of variable stars corresponds to pulsating stars, whose brightness changes periodically due to physical oscillations in their outer layers. In these stars, the equilibrium between gravitational contraction and radiation pressure is regularly disturbed, causing cyclic expansions and contractions. This process leads to variations in luminosity, temperature, and radius that are observable in their light curves. Relevant variables stars belong to this category, such as RR Lyrae, Cepheids, Mira, and delta scuti.

Given the enormous amount of data collected daily from modern observatories, automatic processing and knowledge extraction have become essential

in contemporary astronomy. In this context, machine learning has emerged as a cornerstone methodology for deriving scientific insights from light curves. Techniques such as supervised, unsupervised, and semi-supervised learning have been successfully applied to various tasks, including automatic classification, anomaly detection, and pattern discovery. Over time, these approaches have evolved from hand-crafted feature engineering toward deep learning-based representations, enabling more robust and scalable analyses of astronomical time-series data. Despite these advances, several challenges remain to fully exploit the vast knowledge contained within these massive data sets — for example, the classification of poorly represented objects (e.g. Delta Scuti stars), the processing of signals with low signal-to-noise ratios, and the mitigation of selection and observational biases that can significantly affect model performance and scientific interpretation.

7. Conclusions and further works

Appendix

Data and software availability

The stellar light curves and R codes used in this work are available at <https://github.com/javiersoulforged/JDR>.

Funding

No funds received.

Conflicts of interest/Competing interests

The author declares that they have no known conflict/competing of interest that could have appeared to influence the work reported in this paper.

References

- Akhmetali, A., Zhunuskanov, A., Sakan, A., Zaidyn, M., Namazbayev, T., Turlykozhasheva, D., Ussipov, N. (2025). Luminis Stellarum et Machina: Applications of machine learning in light curve analysis. arXiv preprint arXiv:2504.10038v2.
- Babu, P., Stoica, P. (2010). Spectral analysis of nonuniformly sampled data - a review. Digital Signal Processing 20(2), 359–378.

- Badiane, M., Cunningham, P. (2022). An empirical evaluation of kernels for time series. *Artificial Intelligence Review* 55(3), 1803–1820.
- Bourguignon, S., Carfantan, H., Böhm, T. (2007). SparSpec: a new method for fitting multiple sinusoids with irregularly sampled data. *Astronomy & Astrophysics* 462(1), 379–387.
- Chaini, S., Mahabal, A., Kembhavi, A., Bianco, F.B. (2024). Light curve classification with DistClassiPy: A new distance-based classifier. *Astronomy & Computing* 48, 100850.
- Contreras-Reyes, J.E. (2025). Jensen-distance rate for stationary time series based on cross-spectral methods. *Communications in Nonlinear Science and Numerical Simulation* 149, 108926.
- Cover, T.M., Thomas, J.A. (2006). *Elements of Information Theory*. Wiley & Son, Inc., New York, USA.
- Donoso-Oliva, C., Becker, I., Protopapas, P., Cabrera-Vives, G., Cádiz-Leyton, M., Moreno-Cartagena, D. (2025). Astromer 2: A foundational model for light curve embeddings. *Astronomy & Astrophysics*, accepted (February 2025).
- Dodson-Robinson, S., Haley, C. (2024). Optimal frequency-domain analysis for spacecraft time series: Introducing the missing-data multitaper power spectrum estimator. *The Astronomical Journal* 167(1), 22.
- Duchesne, P., Pacurar, M. (2008). Evaluating financial time series models for irregularly spaced data: a spectral density approach. *Computers & Operations Research* 35(1), 130–155.
- Eyheramendy, S., Elorrieta, F., Palma, W. (2018). An irregular discrete time series model to identify residuals with autocorrelation in astronomical light curves. *Monthly Notices of the Royal Astronomical Society* 481(4), 4311–4322.
- Fotopoulou, S. (2024). A review of unsupervised learning in astronomy. *Astronomy & Computing* 48, 100851.
- Godoy, D.A., Ojeda, C.A. (2025). An Autoregressive Moving Average Model for Time Series with Irregular Time Intervals. *Computer Sciences & Mathematics Forum* 11, 8.

- Huijse, P., Estévez, P.A., Förster, F., Daniel, S.F., Connolly, A.J., Protopapas, P., Carrasco, R., Príncipe, J.C. (2018). Robust period estimation using mutual information for multiband light curves in the synoptic survey era. *Astrophysical Journal Supplement Series* 236(1), 12.
- Kharazmi, O., Contreras-Reyes, J.E., Basirpour, M.B. (2024). Jensen-variance distance measure: A unified framework for statistical and information measures. *Computational and Applied Mathematics* 43(3), 144.
- Mahato, V., O'Reilly, M., Cunningham, P. (2020). A comparison of k-NN methods for time series classification and regression. In *Proceedings of the International Conference on Machine Learning and Data Mining*.
- Ojeda, C., Palma, W., Eyheramendy, S., Elorrieta, F. (2021). A novel first-order autoregressive moving average model to analyze discrete-time series irregularly observed. In *International Conference on Time Series and Forecasting*, pp. 91–103. Cham: Springer International Publishing.
- Ojeda, C., Palma, W., Eyheramendy, S., Elorrieta, F. (2023). Extending time-series models for irregular observational gaps with a moving average structure for astronomical sequences. *RAS Techniques and Instruments* 2(1), 33–44.
- Paparrizos, J., Li, H., Yang, F., Wu, K., D'Hondt, J.E., Papapetrou, O. (2024). A survey on time-series distance measures. *ACM Computing Surveys*, arXiv:2412.20574.
- Piessens, R., deDoncker-Kapenga, E., Uberhuber, C., Kahaner, D. (1983). *Quadpack: a Subroutine Package for Automatic Integration*. Springer-Verlag, Berlin, Germany.
- Rehfeld, K., Marwan, N., Heitzig, J., Kurths, J. (2011). Comparison of correlation analysis techniques for irregularly sampled time series. *Nonlinear Processes in Geophysics* 18(3), 389–404.
- Scargle, J.D. (1989). Studies in astronomical time series analysis. III-Fourier transforms, autocorrelation functions, and cross-correlation functions of unevenly spaced data. *Astrophysical Journal, Part 1* 343, 874–887.
- Schulz, M., Stattegger, K. (1997). SPECTRUM: Spectral analysis of unevenly spaced paleoclimatic time series. *Computers & Geosciences* 23(9), 929–945.

- Springford, A., Eadie, G.M., Thomson, D.J. (2020). Improving the Lomb–Scargle periodogram with the Thomson multitaper. *The Astronomical Journal* 159(5), 205.
- VanderPlas, J.T. (2018). Understanding the Lomb–Scargle periodogram. *The Astrophysical Journal Supplement Series* 236(1), 16.
- Vio, R., Diaz-Trigo, M., Andreani, P. (2013). Irregular time series in astronomy and the use of the Lomb–Scargle periodogram. *Astronomy and Computing* 1, 5–16.
- Wachman, G., Khardon, R., Protopapas, P., Alcock, C.R. (2009). Kernels for periodic time series arising in astronomy. In *Proceedings of the European Conference on Machine Learning (ECML)*, 489–505.
- Xu, K., Qin, M., Sun, F., Wang, Y., Chen, Y.-K., Ren, F. (2021). Learning in the frequency domain. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 1740–1750.